Solution Design Specification (SRS)

Autonomous E-Commerce Operations & Fulfillment Specialist

Target Role: E-Commerce Store & Vendor Ops Specialist

Target Industry: Multi-Vendor E-Commerce Marketplaces

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Evaluation Stage: Level 1 Project (1-Day Assessment)

1. Executive Summary & Problem Breakdown

In modern e-commerce and multi-vendor marketplace platforms, operations teams face critical scaling friction where operational overhead grows linearly with transaction volume. Operations staff spend over 65% of their daily time on repetitive, error-prone manual dashboard updates:

- **Manual Inventory Restock Flagging:** Store operators must manually comb through inventory tables. When a product runs out or dips below minimum safety thresholds, an operator manually flags the item as "RESTOCK NEEDED" on the merchant dashboard and manually drafts emails to suppliers. Inevitably, manual oversights cause stockouts, unfulfillable orders, and lost revenue.
- **Manual Delivery Status Reconciliation:** When third-party couriers (e.g., local dispatch services, DHL, FedEx) complete deliveries, the e-commerce dashboard remains stuck in "Shipped" or "In Transit" until an operations associate manually cross-references courier dispatch reports and clicks "Mark as Delivered." This manual lag delays merchant escrow disbursements, misaligns return windows, and inflates customer service inquiries.

> **Value Proposition:** The Autonomous E-Commerce Operations Specialist operates 24/7 as an intelligent, tool-calling software agent. It eliminates manual data entry, guarantees instant stockout flagging with automated supplier purchase orders, and reconciles courier delivery updates to `DELIVERED` in real-time.

2. Role Breakdown & Prioritization Matrix

The role is decomposed into prioritized operational sub-functionalities:

Workstream / Sub-Functionality	Impact	Complexity	Priority	Included in Level 1
Autonomous Restock Flagging & Inventory Guard	High	Medium	**P0 (Critical)**	Yes
Courier Delivery Status Lifecycle Reconciliation	High	Medium	**P0 (Critical)**	Yes
Gemini 2.5 Flash Tool-Calling Orchestrator	High	Medium	**P0 (Critical)**	Yes
Proactive Background Cron Scheduler	High	Low	**P1 (Core)**	Yes
Transactional Supplier Reorder Email Dispatch	Medium	Low	**P1 (Core)**	Yes
Interactive Full-Stack Operations Dashboard	High	Medium	**P1 (Core)**	Yes
Natural Language Human-in-the-Loop Console	Medium	Medium	**P1 (Core)**	Yes

3. High-Level System Architecture

The solution is designed as a cohesive **npm workspaces monorepo** pairing a reactive operations UI with a tool-calling Node.js backend.



Key Architectural Components:

- **Client (React + Vite + Tailwind CSS):** Renders the real-time operations overview, stockout badges, automated order pipeline transitions, live audit logs, and an interactive AI command console.

- **Backend (Express + TypeScript):** Hosts REST endpoints, courier webhook ingestion, and orchestrates tool calls via Google's Gemini SDK.
- **Agent Core (Gemini 2.5 Flash):** Interprets operational context, reasons over courier proof-of-delivery notes, and autonomously invokes registered system tools.
- **Proactive Scheduler (node-cron):** Executes periodic background audits for delayed shipments and critical inventory depletion without human initiation.

4. Core Agent Tools (Function Calling)

1. `check_inventory_levels()`: Scans SKU balances, identifies items below safety stock, and evaluates restock urgency.
2. `flag_product_for_restock(productId, reason, priority)`: Mutates product flags on the dashboard to `RESTOCK_NEEDED`.
3. `reconcile_delivery_status(orderId, trackingNumber, carrierStatus, deliveryNotes)`: Parses courier delivery proofs and transitions order lifecycle to `DELIVERED`.
4. `dispatch_vendor_restock_email(vendorEmail, sku, quantity, poNotes)`: Dispatches formal supplier purchase orders via transactional email API.
5. `generate_daily_ops_summary()`: Synthesizes operational events, delayed shipment warnings, and restock alerts into an executive health digest.

5. Third-Party Integrations & Free Tier Limits

Service	Role in Solution	Free Tier Limits & Compliance
Google Gemini 2.5 Flash	Autonomous reasoning, intent extraction, courier note analysis, and function calling.	Google AI Studio Free Tier (15 RPM / 1,500 RPD). 100% free, no credit card required.
SendGrid / Resend API	Transactional email dispatch for supplier purchase orders and ops digests.	Free Tier (100 emails/day). 100% free, no credit card required.
Carrier Tracking Webhook Engine	Simulates and ingests 3PL courier tracking feeds and delivery status payloads.	Zero cost, webhook-compatible API simulator adhering to universal logistics standards (17Track / DHL format).

6. Conclusion & Next Steps

This solution directly targets real-world e-commerce operational friction, fulfills all criteria of the Level 1 evaluation, balances frontend usability with robust backend automation, and operates

strictly within zero-cost free-tier boundaries.